

FOUND BUT NOT READ: WHEN EXTRACTED TEXT CLOSES THE RETRIEVAL–READING GAP IN DOCUMENT VISION–LANGUAGE MODELS

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ABSTRACT

Retrieval-augmented document question answering assumes that once the right page is found, a vision–language model (VLM) can read it. We show that this assumption often fails, leaving a retrieval–reading gap: evidence found but not used. A paired protocol isolates this gap by comparing answers from the retrieved page images alone with answers from the same images plus their extracted text. On FoveDoc-Bench, our benchmark with traceable evidence, retrieval finds nearly every evidence page, yet adding CPU-OCR text raises strict accuracy by 13 to 16 points. An exact text layer roughly doubles the gain, which appears across six VLMs from three families and, within one family, narrows with scale without closing. The reader can read this evidence but cannot find it: crops of it recover most of the text gain, boxes around it on the page none. The same protocol identifies two boundaries. Extracted text helps on textual evidence but is neutral or harmful on charts and figures. Its advantage shrinks as retrieval degrades, and unrelated text of the same form adds nothing detectable. Extracted text is an amplifier of retrieval that works, not a substitute for retrieval that does not. Our code is available at <https://github.com/atoz03/fovedoc-sup>.

Index Terms— Document understanding, optical character recognition, vision–language models, retrieval-augmented generation

1. INTRODUCTION

Answering a question about a long document takes two steps: find the pages that hold the evidence, then read them. A retrieval-augmented pipeline [1] gives the first step to a visual retriever [2, 3, 4] and the second to a VLM that receives the top-ranked page images. Progress is tracked by answer accuracy [5, 6, 7] and retrieval recall, and the premise that links them is rarely stated: once the right page is in the context window, reading it is the easy part. Neither number tests the premise, because neither separates a question the retriever lost from one the reader lost.

Our protocol separates them. A page can reach the reader as a pixel array, as in OCR-free models [8, 9], or as a symbolic transcript, as in OCR-augmented ones [10], and a VLM may use evidence in the second form that it fails to extract from the first [11, 12]. The protocol (Fig. 1) holds retrieval at saturation, so nearly every evidence page is in front of the reader, and asks the same question twice: once from the retrieved page images, once from the same images plus a text memory extracted from them. Nothing is removed, so any improvement is evidence found but not used, the *retrieval–reading gap*. Text from a CPU OCR engine, the only text a scanned document can offer, adds 13 to 16 points of strict accuracy; the exact born-digital text layer roughly doubles that. The gap is not a quirk of one backbone: with the text layer it appears in all six stock readers we test from three

Table 1. Benchmarks. Recall@16 is the fraction of questions whose every gold page is among the 16 ColQwen2 retrieves: FoveDoc-Bench is saturated, MMLongBench-Doc is not.

benchmark	questions	pages/doc	gold pages	recall@16
V-NIAH	1173	16–24	1	1.000
V-MQAR	1173	16–24	2	0.998
MMLongBench-Doc	1091*	≤ 468	0– <i>n</i>	0.844

* 135 documents; 838 answerable with annotated evidence, 244 unanswerable.

model families, and within one family it narrows from 2B to 8B parameters without closing. Reading is not the easy part.

The reader can read this evidence; what it cannot do is find it. Crops of the same evidence blocks, with no text at all, recover 87% and 92% of the text gain, while boxes drawn around them on the full pages recover none. Page retrieval leaves a second localization problem inside the context window.

The obvious lesson, OCR every page, is the wrong one, and the same protocol identifies two boundaries of the benefit. Extracted text helps where the evidence is text and is neutral to harmful where it is a chart or figure; on MMLongBench-Doc [7], whose evidence is mostly graphical, the two effects cancel. The advantage also shrinks as retrieval degrades: when we withdraw gold pages from the retrieved set under a fixed page budget, most of the advantage leaves with them, the rest is the document’s own context, and unrelated text of the same form adds nothing detectable. The text does not help because text is easier to consume than pixels; it helps because it delivers evidence the retriever found. Extracted text is an amplifier of retrieval that works, not a substitute for retrieval that does not.

These interventions need evidence that can be traced, withdrawn and re-presented, so we build **FoveDoc-Bench**: 2346 questions over 1173 born-digital documents, each annotated with its evidence pages, blocks and excerpts. We contribute:

1. a **retrieval-controlled paired protocol** that isolates the retrieval–reading gap: 13–16 points with CPU OCR, roughly twice that with an exact text layer, which holds in six readers (Secs. 2–3);
2. an **evidence-accessibility account**: crops recover 87–92% of the text gain and boxes none, and evidence modality and retrieval quality bound the benefit (Secs. 3–5);
3. **FoveDoc-Bench**, traceable evidence for every question, which makes such interventions possible (Sec. 2).

2. THE PAIRED PROTOCOL AND FOVEDOC-BENCH

Arms. ColQwen2 [2] ranks the pages and the top $k=16$ go to the reader, stock Qwen3-VL-2B-Instruct [13] unless stated. The **images** arm receives the 16 pages as rendered images. The **memory** arm receives the same 16 images plus a text memory: the 16 text blocks from those pages that score highest lexically against the question,

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Table 3. MMLongBench-Doc, paired, strict accuracy, same reader, retriever, memory builder and judge as Table 2; Bonferroni $\alpha=0.0125$ over the four strata.

stratum	n	images	+mem.	Δ	p
answerable, annotated	838	0.220	0.223	+0.004	0.86
full recall@16	706	0.227	0.242	+0.016	0.32
partial recall	132	0.182	0.121	-0.061	0.099
unanswerable	244	0.041	0.090	+0.049	0.014

share of the text-layer gain that OCR retains tracks the share of gold evidence that OCR returns intact.

The gap is not a property of one backbone. Fig. 2 repeats the comparison for six stock readers across three families with the text-layer memory: Qwen3-VL-2B/4B/8B [13], Qwen2.5-VL-7B [18], LLaVA-OneVision-7B [19] and Idefics3-8B [20]. The memory arm wins all twelve (reader, task) cells at $p < 10^{-13}$. At 512^2 pixels a page costs 267, 346, 3036 and 3708 image tokens for Qwen3-VL, Qwen2.5-VL, Idefics3 and LLaVA-OV, so the last two read 16 pages with image splitting disabled or pixels capped and gain +0.29 to +0.61 there; the Qwen readers, at native settings, gain +0.19 to +0.36, so across families only the sign is comparable. Within Qwen3-VL, at identical settings, the gap narrows monotonically with scale, from +0.28 to +0.23 on V-MQAR and from +0.36 to +0.23 on V-NIAH between 2B and 8B: a stronger visual reader needs the text less, and at 8B still needs it.

Cheaper levers do not close it. With a 4B reader on V-MQAR, doubling the block budget, a union selector and a reasoning-mode reader each move accuracy by at most +0.03; self-ask decomposition [21] costs -0.043; a 2B LoRA [22] recovers +0.06 on V-MQAR and +0.05 on V-NIAH, and a 4B LoRA adds +0.009. Every lever is small next to the gap.

The reader can read the evidence but cannot find it. Three paired controls locate the failure (V-MQAR / V-NIAH, 1173 samples each). The full OCR text of the 16 pages in reading order, with no question-conditioned selection, gains +0.179 / +0.221, more than the 16-block digest under a neutral prompt (+0.141 / +0.165). The same blocks passed as image crops at the reader’s own page pixel density, appended to the pages without any text, gain +0.123 / +0.152, which is 87% / 92% of the digest’s gain. Red boxes around the same blocks on the page images gain +0.005 / -0.030. The crops carry pixels the pages already held: the reader can read the evidence but cannot find it among 16 dense pages, and lifting it out, as text or as a crop, makes it usable. Page-level retrieval is not evidence-level retrieval.

Takeaway. With every gold page in view, the reader still leaves 13 to 16 points of evidence unused in every backbone we test; lifting it off the page recovers it, with or without question-conditioned selection.

4. THE MODALITY BOUNDARY

MMLongBench-Doc differs from FoveDoc-Bench in the two respects that matter here: its evidence is mostly graphical, and its retrieval is not saturated: full recall@16 is 0.844 (Fig. 3a) and falls with document length, from 0.94 on documents of at most 30 pages to 0.64 above 120. Of the 1091 questions, 244 are deliberately unanswerable, so scoring them measures abstention rather than reading, and 246 carry an empty gold-evidence set; the reading stratum is the 838 questions that are both answerable and annotated, each tagged by the benchmark with the modality of its evidence.

The benefit follows the evidence modality. On the 414 questions carrying exactly one tag (Fig. 3b), the paired effect is +0.085 on text ($n=129$) against -0.046 on chart or figure ($n=285$), a difference of +0.131 [+0.046, +0.215] ($p=0.0024$). The benchmark’s

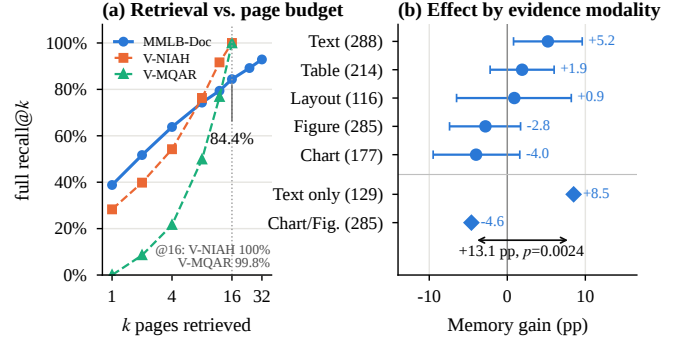


Fig. 3. Retrieval and modality. (a) ColQwen2 full recall@ k : V-NIAH and V-MQAR (1173 each), and MMLongBench-Doc (845 questions with gold pages). (b) MMLongBench-Doc memory – images by evidence tag (tags overlap, 95% CIs) and, below the rule, on the 414 single-tag questions: text against chart/figure differs by +13.1 pp.

own overlapping tags order the same way: plain text +0.052, table +0.019, figure -0.028, chart -0.040. Text evidence gains, as on FoveDoc-Bench; a table, which is text in a grid, sits between; graphical evidence loses, since a flat transcript discards the structure a chart or figure encodes [23, 24].

The two effects cancel in aggregate. Chart and figure evidence weighs 55% of the reading questions against 23% text-only, so the negative stratum carries over twice the weight of the positive one and the pooled advantage is +0.004 (Table 3). The aggregate reports the evidence mix: extracted text is the right representation for textual evidence, and graphical evidence needs one that keeps its structure.

Text also shifts abstention. On the unanswerable stratum the memory arm leads by +0.049 ($p=0.014$, above the corrected α): where the image arm confabulates a number, the memory arm more often reports that the text contains none. Its abstention hit rate rises from 0.029 to 0.086 while its false-alarm rate moves from 0.001 to 0.008, discrimination rather than a bias shift.

Takeaway. Extracted text helps textual evidence and hurts graphical evidence; the 13.1-point contrast marks a modality boundary that an aggregate over a mixed benchmark hides.

5. THE RETRIEVAL BOUNDARY

Table 3 carries a second signal: in the partial-recall stratum the memory arm trails the image arm by 0.061, against a lead of +0.016 where recall is complete. Packed text about the wrong pages misleads more than a page image the reader can decline to attend to, cf. distractor sensitivity in long contexts [25, 26], so retrieval quality should govern the advantage. FoveDoc-Bench’s annotations let us measure how directly, by degrading retrieval under control (Fig. 4).

Design. Retrieval on FoveDoc-Bench is saturated, so we degrade it by hand. **Control 1** withdraws gold pages from the retrieved set and substitutes pages the retriever ranked below the top 16, holding the page budget at 16 so that recall is the only quantity that moves, whereas varying k would confound recall with context length. It scores the same samples at every level, so the pairing is exact twice over, across arms within a level and across levels within a sample, and the change in the advantage between full and zero recall is a within-sample interaction: the part of the advantage that the retrieved gold pages buy. **Control 2** restricts zero recall to samples whose packed memory does not contain the answer string, detected by normalised string containment at four minimum answer lengths, since values recur in abstracts, headers and cross-references, so that the residual is

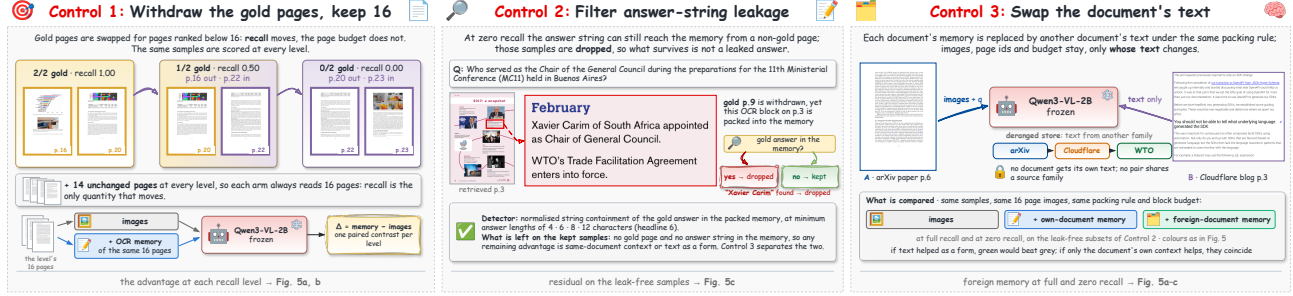


Fig. 4. Three controls on the memory advantage, all with OCR memory, each on a real sample; results in Fig. 5. **Control 1** varies recall at a fixed 16-page budget on the same samples. **Control 2** drops zero-recall samples whose packed memory contains the answer string. **Control 3** builds the memory from another document's text under the same rule.

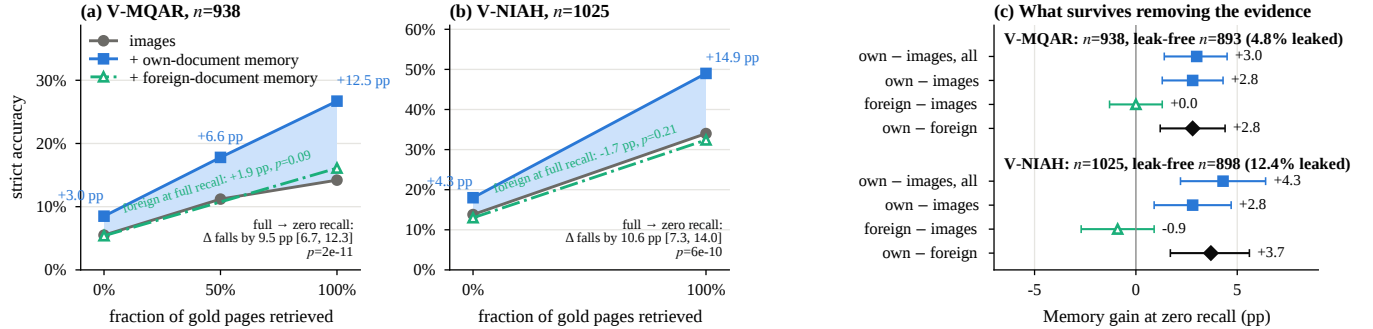


Fig. 5. Results of the controls in Fig. 4. **(a, b)** The memory advantage (labelled) falls monotonically as gold pages are withdrawn; foreign-document text tracks the image arm at both endpoints. **(c)** At zero recall the leak-free residual survives, but foreign-document memory of identical form is indistinguishable from the image arm: what remains is same-document context, not format. Error bars are 95% CIs.

not a leaked answer. **Control 3** replaces each document's memory with a *different* document's text under the same packing rule and budget; the assignment is deranged, no document receiving its own text and no pair sharing a source family, so foreign memory can neither leak the answer nor be topically related, and it isolates what a text memory is worth for its form alone. Eligible samples, whose documents can supply substitutes at every level, number 938 for V-MQAR (two gold pages, three levels) and 1025 for V-NIAH (one gold page, two levels).

Retrieved evidence buys most of the advantage. The advantage falls monotonically as gold pages are withdrawn (Fig. 5a,b), from +0.125 through +0.066 at half recall to +0.030 on V-MQAR and from +0.149 to +0.043 on V-NIAH. The within-sample change is +0.095 [+0.067, +0.123] and +0.106 [+0.073, +0.140], both $p < 10^{-9}$: roughly three quarters of the full-recall advantage is bought by the gold pages being present. The residual at zero recall is small but not zero, and part of it is leakage: the gold answer string sits in the packed zero-recall memory of 4.8% of V-MQAR samples and 12.4% of V-NIAH samples. On the leak-free subsets ($n=893$ and 898) the residual is +0.028 on both tasks ($p=6 \times 10^{-4}$ and 6×10^{-3}), and it stays between +0.025 and +0.031 as the detector's minimum answer length varies from 4 to 12 characters.

The residual is context, not format. Foreign memory at zero recall is indistinguishable from the image arm on both tasks, +0.000 [-0.013, +0.013] ($p=0.87$) and -0.009 [-0.027, +0.009] ($p=0.40$), and own - foreign is the whole of the residual, +0.028 [+0.012, +0.044] and +0.037 [+0.017, +0.056], both $p < 10^{-3}$: what survives is the document's own non-gold text.

The control also speaks to Sec. 4. Even at *full* recall foreign text is inert rather than harmful (+0.019 and -0.017, both $p > 0.08$): a reader handed a page of unrelated prose beside the right

images simply ignores it. Contrast the chart-and-figure stratum, where memory of the *correct* pages actively hurt: what damages the reader is relevant text about the wrong modality, not irrelevant text.

Takeaway. Three quarters of the advantage is bought by retrieved evidence and the rest by the document's own context; text as a format, without that content, is worth nothing.

6. CONCLUSION

Under a retrieval-controlled paired protocol, vision-language readers leave evidence unused on pages they already hold. Extracted text recovers much of it: CPU OCR adds 13 to 16 points of strict accuracy over the page images alone, an exact text layer roughly doubles that, and with the text layer the gain appears in all six readers we test across three model families. The reader can read the evidence it cannot find: crops of it recover most of the text gain, boxes around it on the page none. Two boundaries govern the benefit. The evidence must be textual: on charts and figures extracted text is neutral to harmful, and on a mostly graphical benchmark the two effects cancel. And retrieval must work: three quarters of the advantage is bought by the gold pages being present, the rest is the document's own context, and unrelated text of the same form is worth nothing detectable. An extracted-text memory is an amplifier of retrieval that works, not a substitute for retrieval that does not.

Two practical conclusions follow. An OCR-then-read stage belongs in text-evidenced document QA over strong retrieval, gated on retrieval quality and evidence modality rather than applied to every page. And evaluations of such pipelines should be stratified by recall and modality, since an aggregate over one mix does not transfer: the same intervention measures +0.13 on FoveDoc-Bench and +0.004 on MMLongBench-Doc, and both numbers are right.

7. COMPLIANCE WITH ETHICAL STANDARDS

This is a computational study on publicly available documents and benchmarks. It involved no human or animal subjects, and no ethical approval was required.

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